

VisualRAG: Knowledge-Guided Retrieval Augmentation for Image-Text Matching

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Abstract—Image-text matching as a fundamental cross-modal understanding task presents unique challenges in weakly-aligned scenarios. Such data typically feature highly abstract textual captions with sparse entity References, creating a significant semantic gap with visual content. Current mainstream methods, primarily designed for strongly aligned data pairs, employ dynamic modeling or multi-dimensional similarity computation to achieve feature space mapping. However, they struggle with information asymmetry and modal heterogeneity in weakly aligned cases. To address this, we propose a Visual Perception Knowledge Enhancement (VPKE) framework. Unlike existing methods based on strong alignment assumptions, this framework mines latent image semantics through vision-language models and generates auxiliary captions, overcoming the information bottleneck of traditional text modalities. Its core innovation lies in an adaptive knowledge distillation mechanism that combines retrieval-augmented generation (RAG) with key entity extraction. This mechanism effectively filters noise when introducing external knowledge while optimizing cross-modal feature integration. The framework employs multi-level similarity evaluation to dynamically adjust fusion weights among original text, key entities, and auxiliary captions, enabling adaptive integration of diverse semantic features and significantly improving model flexibility. Additionally, multi-scale feature extraction further enhances cross-modal representation capabilities. Experimental results show that the proposed method performs excellently in image-text retrieval tasks on the MSCOCO and Flickr30K datasets, validating its effectiveness.

Index Terms—Image-text matching, knowledge enhancement, large language model, modality heterogeneity.

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I. INTRODUCTION

WITH the rapid development of internet technology, the ways in which information is disseminated have become increasingly diverse. Image-text information has become an important medium for people to acquire and exchange knowledge. Therefore, cross-modal research [1], [2], [3], [4] has gained widespread attention in academia in recent years. Image-text matching, as a key research direction, aims to effectively align different modalities. Specifically, this involves identifying the corresponding image through a text query, or vice versa.

Current research [5], [6], [7], [8], [9] primarily focuses on addressing the modality gap problem in vision-language matching, aiming to improve semantic alignment between images and texts. As illustrated in Fig. 1(a), visual content and linguistic captions exhibit significant distribution discrepancies in the feature space, where this semantic heterogeneity constitutes the core challenge of cross-modal matching. The semantic similarity learning framework proposed by Zhang et al. [10] enhances modal alignment by dynamically modeling the relationship between visual and textual features. However, existing methods [11], [12], [13], [14] generally suffer from two limitations. First, they overly rely on inter-modal relationships while neglecting in-depth exploration of cross-modal features. Second, they fail to effectively preserve original data details, particularly the weak correspondence between image objects and textual entities, leading to inconsistent modal representations. To address this issue, researchers have gradually shifted toward leveraging large-scale vision-language models (such as CLIP [15] and BLIP [16]) and prior knowledge generated by AIGC techniques to construct semantic bridging paths. As shown in Fig. 1(b), auxiliary captions can establish semantic bridging paths to reduce modal heterogeneity. Nevertheless, these methods suffer from rigid knowledge integration, lacking dynamic optimization of key information and adaptive fusion mechanisms, which restricts their practical application performance. Taking BLIP as an example, despite its strong generative capability, inherent flaws impair cross-modal matching. The generated auxiliary captions often contain irrelevant visual details, mismatching key textual semantics, and the single-pass generation process fails to effectively filter out task-irrelevant entities. A deeper challenge lies in simultaneously resolving modality heterogeneity and information completeness gaps. Modality heterogeneity

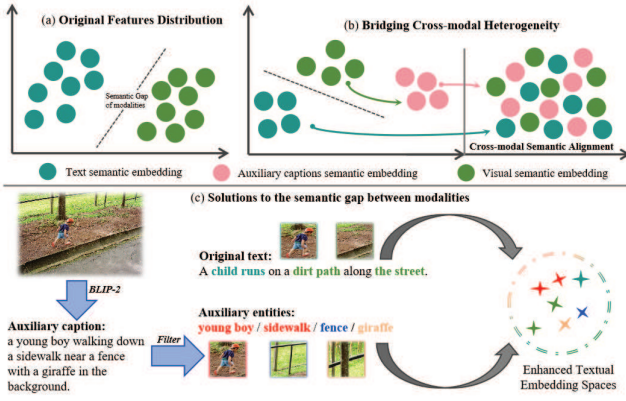


Fig. 1. Heterogeneity and Semantic Gap in Image-Text Retrieval and Our Proposed Solutions. (a) visually illustrates the semantic heterogeneity between images and texts; (b) employs auxiliary captions to construct semantic bridging paths for mitigating modal heterogeneity. (c) demonstrates asymmetric information across modalities and addresses semantic information disparities through generated captions.

arises from the distributional divergence between image and text data in feature space, where these modalities exhibit distinct statistical properties or semantic structures. The inter-modal information discrepancy refers to variations in semantic completeness, where images may contain fine-grained visual details while their corresponding text descriptions provide only abstract summaries. Although existing models [17], [18], [19] have made progress, they still face significant limitations in weakly aligned scenarios. For instance, in the content recommendation [20] of social media, associating ambiguous texts (e.g., “hanging out with friends”) with visual elements proves difficult, leading to recommendation bias. In autonomous driving [21], [22], misalignment between abstract instructions (e.g., “obstacle ahead”) and point cloud data threatens system reliability. These cases reveal the positive correlation between text information density and retrieval accuracy, as well as the difficulty of achieving strongly aligned semantic equivalence. Notably, as depicted in Fig. 1(c), while generated captions can alleviate information asymmetry, models like BLIP suffer from excessive generation of irrelevant details and lack selective entity filtering. Therefore, establishing dynamic feature optimization mechanisms and adaptive fusion strategies becomes a critical pathway to overcoming existing technical bottlenecks and enhancing cross-modal task performance. This requires addressing the dual challenges of feature space distribution differences and semantic completeness gaps simultaneously, achieving true semantic equivalence alignment through more refined modal interaction designs.

To address these challenges, we recognize the critical importance of uncovering latent visual and textual information in images while incorporating finer-grained details to facilitate image-text matching. BLIP-2 [23], as a zero-shot vision-language model, can extract rich visual information from images, such as “young boy,” “sidewalk,” “giraffe,” and “fence,” while potentially introducing irrelevant entities that conflict with the original semantics (e.g., “background,” “near,” and “walking down”), as shown in Fig. 1(c). To mitigate this, we propose a visual perception knowledge

enhancement framework based on the BLIP-2 architecture, which aims to bridge modal discrepancies, enrich semantic representations, and resolve potential conflicts by incorporating external knowledge sources. Specifically, to tackle modal information disparities, we design a cross-modal knowledge retrieval enhancement module. This module leverages vision-language models to extract latent visual features from images and generate auxiliary captions, thereby compensating for the sparsity and abstraction of textual semantics while ensuring relevance and accuracy through conflict-aware filtering. By integrating retrieval-augmented generation (RAG) [24] with key entity extraction techniques, the module effectively filters out noisy or conflicting entities while preserving essential cross-modal features through dual-confidence evaluation. This process enhances semantic information in the textual modality and expands the number of coherent entities. To address modal heterogeneity and conflicts between auxiliary captions and original caption texts, we further propose a similarity-based hierarchical vision-guided semantic rebalancing module. This module dynamically adjusts the contribution weights among original texts, key entities, and auxiliary captions through conflict-sensitive multi-metric similarity evaluation, enabling adaptive fusion of heterogeneous semantic features by suppressing conflicting elements. Finally, we employ a multi-scale hierarchical attention aggregator module to extract multi-scale features from images and texts, strengthening the semantic representation capability for cross-modal learning. To fully utilize the feature extraction power of pretrained models, we adopt image and text encoders from vision-language models like CLIP, ensuring reliable and robust cross-modal feature representation. The key contributions are as follows:

- We propose a cross-modal knowledge retrieval enhancement module (CKRE) that utilizes large-scale vision-language models to extract visual features and generate auxiliary captions. This module innovatively integrates the RAG strategy to build a knowledge base for key entity extraction, effectively filtering noise while retaining essential features.
- We propose a similarity-based vision-guided semantic rebalancing (VR) model that dynamically adjusts the fusion weights of original textual features, key entity features, and auxiliary caption features through multi-dimensional semantic evaluation. This approach effectively integrates heterogeneous semantics and enhances representation robustness. Additionally, we employ a multi-scale hierarchical attention aggregator (MHA) module to extract multi-scale features, further improving semantic expression in cross-modal learning.
- Extensive experiments on Flickr30K and MSCOCO demonstrate state-of-the-art performance of our method in image-text retrieval. Our code is available at https://github.com/hygge-w/VPKE_mian.

II. RELATED WORK

A. Image-Text Matching

To reduce the heterogeneity gap between modalities, researchers [25], [26], [27], [28] have conducted extensive

studies. Recent advances in image-text retrieval are introduced from two perspectives.

1) From the perspective of global alignment, existing methods [29], [30], [31], [32] map image and text features into a shared space but lack cross-modal feature interaction. Zhang et al. [33] introduce a semantic optimization method for image-text retrieval that enhances shared space mapping by employing visual semantic loss to mitigate minor content interference and emphasize primary image features. Xu et al. [34] propose a Transformer-based network that enhances global semantics and integrates multi-scale data via a confidence-guided probabilistic module, maintaining cross-domain consistency while reducing modal interference.

2) In the realm of fine-grained alignment [5], [10], [35], [36], there is a general emphasis on leveraging specific, localized features to calculate similarity. Pang et al. [37] propose TVRN, integrating visual context into text to reduce semantic ambiguity by combining local and global similarities. Zhang et al. [38] develop LG-MGC, enhancing cross-modal retrieval via local semantic completion and generative transformation. Zhang et al. [39] introduces causal inference to identify crucial semantic segments through dual-path modeling. Li et al. [40] propose a memory-enhanced framework based on cross-modal correlation graphs to achieve fine-grained interaction. Xiong et al. [41] propose a reference attention module to refine cross-modal relationships by reducing incorrect attention scores. Although fine-grained methods achieve better performance, they require higher computational costs. Current image-text matching methods overly rely on modal relationships while neglecting real-time semantic interaction and dynamic correction. Moreover, insufficient textual representation details hinder alignment performance. This paper constructs a visually guided cross-modal knowledge feedback loop to mine image features and supplement textual semantics, enabling coarse-to-fine feature collaboration.

B. Knowledge-Enhanced Cross-Modal Representation

Although deep learning [42], [43], [44], [45] has made significant progress in various fields, it still faces challenges such as heavy dependence on data, limited interpretability, and instability when deployed in unfamiliar environments. In this context, the incorporation of prior knowledge can effectively address these issues, especially in cross-modal tasks. By injecting external knowledge, models can improve their performance and enhance their ability to capture intricate details. Current mainstream approaches for multimodal knowledge enhancement primarily rely on large-scale vision-language models (e.g., CLIP, BLIP) and AIGC technologies to generate additional modal knowledge and integrate it into the model. For instance, Wu et al. [9] propose Cap4Video, which generates video captions via zero-shot video description by combining CLIP and GPT-2 [46], thereby enriching training data and improving text-video retrieval. Pu et al. [47] introduce MiC, which enhances multimodal consistency by comparing original text-visual pairs with generated unimodal knowledge. Tschannen et al. [48] present CLIPPO, rendering text as images and processing them with a unified ViT, eliminating modality-specific encoders. Dong et al. [25] propose

TABLE I
SOME IMPORTANT ABBREVIATIONS AND NOTATIONS

Abbreviation and Notation	Definition
$T = \{t_i \mid i \in [1, 2, \dots, n], t_i \in \mathbb{R}^d\}$	Textual features
$I = \{v_i \mid i \in [1, 2, \dots, n], v_i \in \mathbb{R}^d\}$	Image (visual) features
$\bar{T} = \{\bar{t}_i \mid i \in [1, 2, \dots, n], \bar{t}_i \in \mathbb{R}^d\}$	Auxiliary caption features
$C = \{c_i \mid i \in [1, 2, \dots, n^k], c_i \in \mathbb{R}^d\}$	Key entity features
$T^f = \{t_i^f \mid i \in [1, 2, \dots, n], t_i^f \in \mathbb{R}^d\}$	Generated fusion textual features
$T^c = \{t_i^c \mid i \in [1, 2, \dots, n], t_i^c \in \mathbb{R}^d\}$	Entity-enriched textual features
$T^{cl} = \{t_i^{cl} \mid i \in [1, 2, \dots, n], t_i^{cl} \in \mathbb{R}^d\}$	Multi-scale entity-enriched textual embeddings
$T^{fl} = \{t_i^{fl} \mid i \in [1, 2, \dots, n], t_i^{fl} \in \mathbb{R}^d\}$	Multi-scale generated fusion textual embeddings
$I^l = \{v_i^l \mid i \in [1, 2, \dots, n], v_i^l \in \mathbb{R}^d\}$	Multi-scale visual embeddings
$P = \{p_i\}_{i=1}^n$	A phrase set for each auxiliary caption.
M_D	Concept memory bank constructed from the training set

I-CLIP, enriching text modality in weakly aligned settings using an imagination network and optimizing semantics via reinforcement learning. Unlike these methods, we pioneer the combination of RAG and key entity extraction to refine external knowledge integration, enabling noise filtering and collaborative optimization of cross-modal features. Through multi-level similarity evaluation, we dynamically adjust the weights of text, entities, and auxiliary captions, enhancing adaptive fusion.

III. METHODOLOGY

To address the image-text matching problem from the perspective of information disparity and weak alignment, we propose a groundbreaking visual-perception knowledge enhancement framework (VPKE), as shown in Fig. 2. It consists of four key modules: the CLIP encoder (Sec. III-A), the cross-modal knowledge retrieval enhancement module (Sec. III-B), the vision-guided semantic rebalancing mechanism (Sec. III-C), and the multi-scale hierarchical attention aggregator module (Sec. III-D). In this section, we focus on the function of the agent, particularly how it generates richer and more comprehensive textual entities through the collaboration of multiple modalities, thereby increasing the information content of the text modality.

A. Modality Encoders

Symbol Set Definition: The dataset D contains two modalities, images and texts, represented by IMG and TXT , respectively. Let $D = \{D_i\}_{i=1}^n$, where n denotes the number of image-text pairs. Each $D_i = (IMG_i, TXT_i)$, and IMG_i and TXT_i are the original image and text of the i -th instance, respectively. The collection of textual features is denoted by $T = \{t_i \mid i \in [1, 2, \dots, n], t_i \in \mathbb{R}^d\}$ and collection of image (visual) features is represented by $I = \{v_i \mid i \in [1, 2, \dots, n], v_i \in \mathbb{R}^d\}$. Here, d is the dimension of the feature vectors. In the text, bold lowercase letters represent vectors, ordinary lowercase letters are used to denote scalars, and bold uppercase letters are employed to indicate matrices composed of multiple vectors. The definitions of key symbols are detailed in Table. I

Within our model framework, the initial features of the image and text for the i -th sample are obtained by leveraging the corresponding encoders of the CLIP model, parameterized

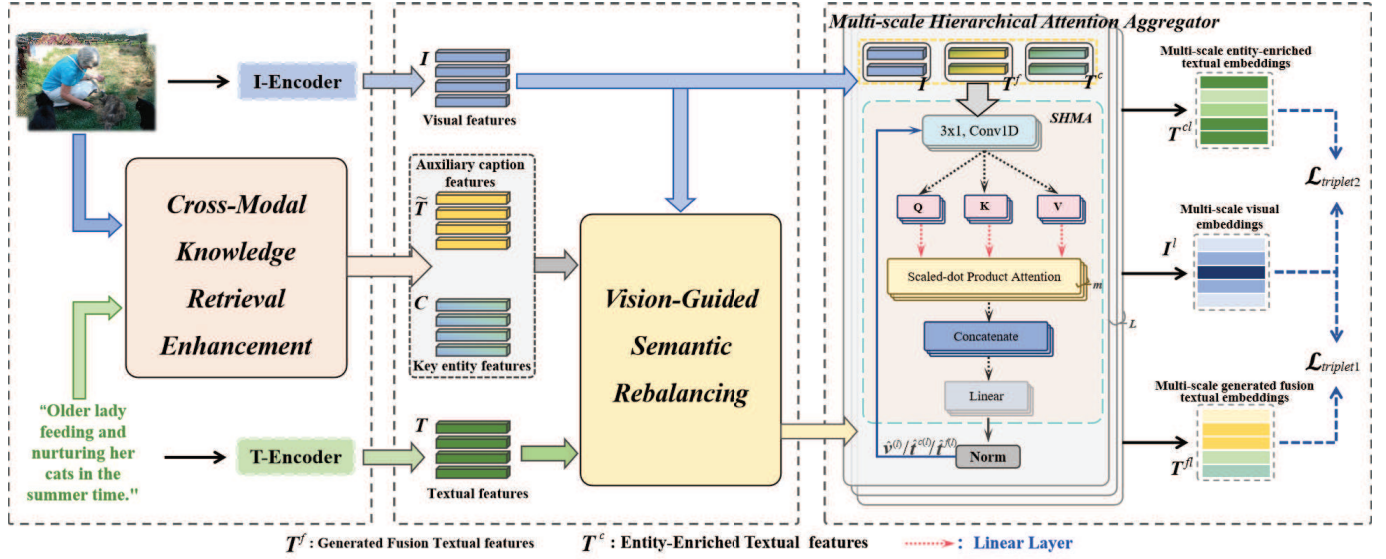


Fig. 2. Illustration of our proposed VPKE. The cross-modal knowledge retrieval enhancement module mines image semantics to optimise cross-modal features. Multi-level similarity evaluation dynamically adjusts the fusion strategy, thereby achieving adaptive fusion and enhancement. The multi-scale hierarchical attention aggregator module extracts features at multiple scales in order to strengthen semantic representation and improve performance.

as θ_T and θ_I . These features are denoted as v_i for the image and t_i for the text, respectively.

$$\begin{aligned} t_i &= T_{\text{encoder}}(TXT_i; \theta_T), \\ v_i &= I_{\text{encoder}}(IMG_i; \theta_I). \end{aligned} \quad (1)$$

In image-text pairs, we design a cross-modal knowledge retrieval enhancement module that effectively promotes the interaction between textual and visual features in tasks involving semantically rich image-text alignment. This module enhances the robustness of feature representations, contributing to more accurate similarity matching.

B. Cross-Modal Knowledge Retrieval Enhancement

Existing cross-modal retrieval methods [49], [50] face a critical challenge: the inherent sparsity of textual modality information. When text captions contain limited entities, their semantic alignment with image content is significantly hindered by insufficient representation dimensions. This fundamental limitation severely restricts the potential improvement of image-text matching accuracy. Guided by cross-modal semantic completeness theory, we propose a novel cross-modal knowledge retrieval enhancement module, as shown in Fig. 3. This module consists of an image caption generator and an adaptive knowledge refinement strategy, establishing a semantic compensation mechanism from visual to textual modalities. While the BLIP-2 model extracts latent image semantics and generates auxiliary descriptive text to expand textual representation dimensions, it inevitably introduces noise unrelated to the image. Such noise not only disrupts cross-modal feature alignment but may also cause semantic drift. To address this, we design an adaptive knowledge refinement module that integrates retrieval-augmented generation with key entity extraction. During knowledge fusion, this module simultaneously filters noise and optimizes features. It intelligently identifies and retains semantics highly relevant to

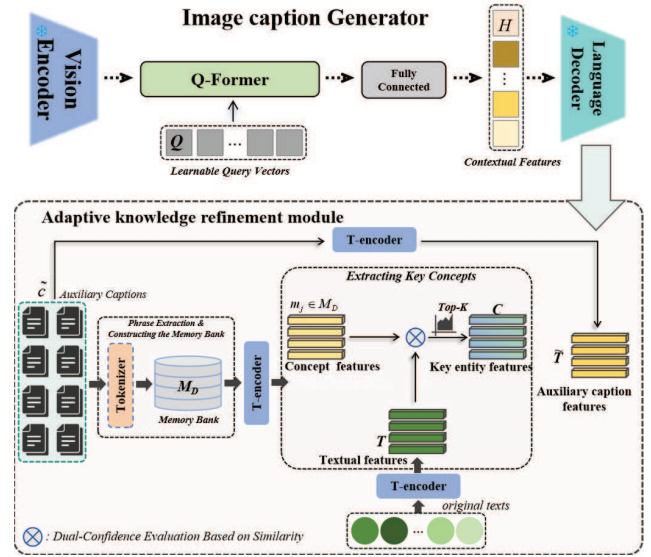


Fig. 3. Cross-Modal Knowledge Retrieval Enhancement Module. First, an image caption generator is used to produce auxiliary captions. Then, through part-of-speech analysis and filtering processes, an external knowledge memory bank is constructed. Next, by combining RAG and key entity extraction techniques, key entities are selected from the knowledge memory bank M_D .

the image while effectively eliminating noise from auxiliary captions.

1) *Image Caption Generator*: BLIP-2 is a zero-shot vision-language model designed for various image-to-text tasks involving image and text prompts. It is a highly effective and efficient approach applicable to diverse scenarios in image understanding, particularly when training samples are scarce. The model bridges the gap between visual and natural language modalities by incorporating transformers into the pre-trained models. This novel pre-training paradigm allows it to fully benefit from advancements in both modalities. First,

given an input image IMG_i , the vision encoder first extracts hierarchical features v_i :

$$v_i = \text{VisionEncoder}(IMG_i) \in \mathbb{R}^{N_v \times d_v}, \quad (2)$$

where N_v denotes the number of visual tokens, representing the number of visual units in the image, d_v denotes the feature dimension, representing the feature length of each visual token. Next, learnable query vectors $Q \in \mathbb{R}^{N_q \times d_v}$ interact with the visual features through cross-attention to generate contextual features:

$$H_i = \text{Q-Former}(Q, v_i) \in \mathbb{R}^{N_q \times d_v}, \quad (3)$$

where N_q denotes the number of query tokens, and H_i are the generated contextual features, with dimensions consistent with the query vectors. Finally, the contextual features are input into the language decoder to generate the auxiliary captions $\tilde{c} = \{\tilde{c}_i \mid i \in [1, 2, \dots, n], \tilde{c}_i = \{w_t\}_{t=1}^{N_m}\}$. The generation process is modeled as:

$$P(\tilde{c}_i | IMG_i) = \prod_{t=1}^m P(w_t | w_{<t}, \text{Proj}(H_i)), \quad (4)$$

where $P(w_t | w_{<t}, \text{Proj}(H_i))$ represents the probability of generating the current word w_t given the first $t-1$ words and the contextual features; N_m is the total number of words in the generated text (text length); t indicates the position index of the current word in the text; $\text{Proj}(\cdot)$ is a dimension alignment projection layer, used to map the contextual features to the dimension required by the language model. Subsequently, $\tilde{c}$ is encoded into the auxiliary caption features $\tilde{T} = \{\tilde{t}_i \mid i \in [1, 2, \dots, n], \tilde{t}_i \in \mathbb{R}^d\}$ using Eq. (1).

2) Adaptive Knowledge Refinement: Through the above processing flow, the system generates new auxiliary captions information for each image-text pair. It is important to note that pre-trained vision-language models often suffer from hallucination issues during generation, easily introducing irrelevant noise. To avoid interference from such noise in model training, the candidate vocabulary range must be strictly controlled. To achieve this, the module employs contextual semantic analysis, part-of-speech rules, and stop-word filtering techniques to effectively remove unrelated scene entity captions. This ensures that only relevant semantic information is supplemented. The module accurately retains key entity terms while extracting semantically rich phrase combinations, enhancing text representation while ensuring captions accuracy.

Phrase Extraction: Given an auxiliary caption, it is first segmented into individual words or phrases. Then, each word or phrase is tagged with its corresponding part-of-speech (POS). This process produces a tagged sequence of $\{(w_1, a_1), (w_2, a_2), \dots, (w_n, a_n)\}$, where a_i represents the POS tag associated with the word or phrase w_i . To extract meaningful phrases and words, they are defined as continuous subsequences of tags. The phrase set P is iteratively constructed by filtering whether each tag in this subsequence belongs to the set $O = \{\text{NN}, \text{NNS}, \text{NNP}, \text{NNPS}, \text{JJ}\}$ of POS tags. If a non-matching POS tag is encountered, the

current phrase or word is removed from consideration. The formalization of this process is as follows:

$$P = \begin{cases} \{(w_i, a_i)\} & \text{if } a_i \in O, \\ P \cup \{(w_{i+1}, a_{i+1})\} & \text{if } a_{i+1} \in O, \\ \text{end} & \text{if } a_{i+1} \notin O, \end{cases} \quad (5)$$

where P represents the complete set of phrases stored for each auxiliary captions, NN denotes singular nouns, NNS represents plural nouns, NNP indicates singular proper nouns, NNPS refers to plural proper nouns, and JJ signifies adjectives. These tags assist in syntactic analysis and semantic understanding of texts, providing foundational support for tasks such as information extraction and natural language generation. For each phrase set P containing phrases or words $\{p_i\}_{i=1}^n$, traverse each element $p_i \in P$ and calculate the token count $|p_i|$. If there exists a phrase $p_i \in P$ such that $|p_i| > 1$, meaning the element consists of more than one token, it is directly added to the key concept set M_I . For single-token words $p_i \in P$ with $|p_i| = 1$, further check whether the token p_i exists in the stopword set S^t or the punctuation set S^p . If the token $p_i \notin S^t \cup S^p$, it is also added to M_I . This process is as follows:

$$M_I = M_I \cup_{p_i \in P} \begin{cases} p_i, & \text{if } |p_i| > 1, \\ p_i, & \text{if } |p_i| = 1 \wedge p_i \notin S^t \cup S^p. \\ \emptyset, & \text{otherwise.} \end{cases} \quad (6)$$

Constructing the Memory Bank: We apply the above extraction method to the auxiliary captions of all images in dataset D , focusing only on training-set-labeled images. For each auxiliary caption, we follow the previously outlined extraction process. By extracting phrases from all auxiliary captions, we construct a key concept set M_I for each image. These sets are then aggregated across the entire training set to form a comprehensive concept memory bank M_D , which functions as the external knowledge base:

$$M_D = \bigcup_{I \in D_{\text{train}}} M_I, \quad (7)$$

where M_D consists solely of data from the training set.

Extracting Key Concepts: We propose a feature selection mechanism based on dual confidence evaluation. By calculating the deep semantic similarity s_{ij} between the original textual feature t_i and each concept feature $m_j \in M_D$, a confidence-weighted similarity matrix is constructed. The confidence score g_{ij} is derived by evaluating semantic consistency through a lightweight sigmoid-linear transformation of the concatenated features:

$$g_{ij} = \sigma(\mathbf{w}_m^T [t_i; m_j; s_{ij}] + b), \quad (8)$$

where σ is the sigmoid function and $[\cdot; \cdot]$ denotes vector concatenation, $\mathbf{w}_m \in \mathbb{R}^{2d+1}$ is the weight vector directly mapped to the confidence, and b is the bias. The final comprehensive score is $\mathbf{r}_{ij} = s_{ij} \cdot g_{ij}$, and the top-k features are selected by sorting $\mathbf{r}_{ij}$. This method adaptively selects the top-k key entity features $\mathbf{C} = \{c_i \mid i \in [1, 2, \dots, n^k], c_i \in \mathbb{R}^d\}$ with the highest comprehensive scores. n^k represents that each key entity feature is composed of G_k key concept words. This feature distillation approach ensures the reliability of feature selection through the dual confidence mechanism, effectively improving model performance in complex semantic understanding tasks.

C. Vision-Guided Semantic Rebalancing

Vision-Guided Semantic Rebalancing module develops a visual-guided dynamic feature fusion approach to address semantic conflicts and modality heterogeneity in multimodal tasks. This method synergistically integrates auxiliary captions, which provides global semantic context, with key entities that anchor high-confidence visual concepts, thereby complementarily enhancing semantic completeness and robustness. As illustrated in Fig. 1(c), while accurately identifying visual elements such as “giraffe” and “fence” in images, the generated auxiliary captions may contain irrelevant entities conflicting with original semantics. For instance, when the actual image action is “runs”, the generated auxiliary caption might erroneously describe it as “walking down”. Conventional static fusion methods indiscriminately incorporate such erroneous information, resulting in semantic drift. To overcome this limitation, this module implements a multi-level similarity evaluation mechanism that dynamically adjusts the weight distribution among textual features, auxiliary caption features, and key entity features, effectively suppressing interference from incorrect semantics.

First, the average cosine distance between each text feature and the visual feature is calculated as a similarity measure:

$$\begin{aligned} x_1 &= \text{Cos}(\mathbf{T}, \mathbf{I}) = \frac{1}{n} \sum_{i=1}^n \left(1 - \frac{\mathbf{t}_i \cdot \mathbf{v}_i}{\|\mathbf{t}_i\| \|\mathbf{v}_i\|} \right), \\ x_2 &= \text{Cos}(\tilde{\mathbf{T}}, \mathbf{I}) = \frac{1}{n} \sum_{i=1}^n \left(1 - \frac{\tilde{\mathbf{t}}_i \cdot \mathbf{v}_i}{\|\tilde{\mathbf{t}}_i\| \|\mathbf{v}_i\|} \right), \\ x_3 &= \text{Cos}(\mathbf{C}, \mathbf{I}) = \frac{1}{n} \sum_{i=1}^n \left(1 - \frac{\mathbf{c}_i \cdot \mathbf{v}_i}{\|\mathbf{c}_i\| \|\mathbf{v}_i\|} \right), \end{aligned} \quad (9)$$

where x_1 , x_2 , and x_3 respectively quantify the overall similarity between the original textual features, the auxiliary caption features, and the key entity features with the current visual features. Based on the above similarity values, the corresponding weights are calculated: $w_1 = x_1 / (x_1 + x_2)$, $w_2 = x_2 / (x_1 + x_2)$, $w_3 = x_1 / (x_1 + x_3)$ and $w_4 = x_3 / (x_1 + x_3)$. w_1 and w_2 are used to balance the contributions of the textual features and the auxiliary caption features, whereas w_3 and w_4 are responsible for regulating the fusion ratio between the textual features and the key entity features. These weights are dynamically adjusted under the guidance of visual features, aiming to highlight key content and suppress noise interference. Ultimately, the fused features are generated progressively through the following two pathways:

$$\begin{aligned} \mathbf{T}^f &= w_1(w_1\mathbf{T} + w_2\tilde{\mathbf{T}}) + w_2\tilde{\mathbf{T}}, \\ \mathbf{T}^c &= w_3(w_3\mathbf{T} + w_4\mathbf{C}) + w_4\mathbf{C}, \end{aligned} \quad (10)$$

where $\mathbf{T}^f = \{\mathbf{t}_i^f \mid i \in [1, 2, \dots, n], \mathbf{t}_i^f \in \mathbb{R}^d\}$ denotes the generated fusion textual features that incorporate the auxiliary caption features, while $\mathbf{T}^c = \{\mathbf{t}_i^c \mid i \in [1, 2, \dots, n], \mathbf{t}_i^c \in \mathbb{R}^d\}$ represents the entity-enriched textual features resulting from the fusion with key entity features. This dual-weighted fusion mechanism achieves dynamic feature integration through progressive nonlinear interactions. Under visual guidance, it adaptively balances the synergistic and competitive relationships

among multimodal features. Consequently, it effectively suppresses noise interference while highlighting crucial semantic information, thereby significantly enhancing the robustness and generalization performance of image-text matching.

D. Multi-Scale Hierarchical Attention Aggregator

To achieve richer information granularity, we propose a multi-scale hierarchical attention aggregator that integrates multi-layer convolution with a spatial attention-guided hierarchical multi-head attention mechanism (SHMA) for multi-scale extraction of image and text features, followed by similarity computation based on these features.

1) *Hierarchical Processing*: The module aims to extract multi-scale features from both image and text to enhance their semantic expressiveness in cross-modal learning. Taking the processing of image features $\mathbf{v}$ as an example, the following is a detailed description of the method. Specifically, it gradually refines image features through a series of convolutional layers and the SHMA mechanism. Multiple convolutional kernels are used to hierarchically process the image embeddings, while the SHMA mechanism further enhances the feature representation. First, for the input image features, a parameter-sharing parallel convolutional architecture is employed for multi-scale feature learning:

$$\mathbf{v}^{(l)} = \sigma \left(\mathbf{W}_C^{(l)} * \mathbf{v}^{(l-1)} + \mathbf{b}^{(l)} \right), \quad l = 1, \dots, L, \quad (11)$$

where $\mathbf{v}^{(l)}$ denotes the convolutional output features at the l -th layer, $(*)$ denotes the depthwise separable convolution operation, $\mathbf{W}_C^{(l)}$ represents the learnable convolution kernel at the l -th layer, and $\mathbf{b}^{(l)}$ denotes the learnable bias term at the l -th layer. Additionally, $\sigma(\cdot)$ is the GeLU activation function. By dynamically adjusting the dilation rate of the convolution kernels, a series of feature representations at different scales are constructed. After multi-scale feature learning, the SHMA mechanism is applied to further enhance the feature representation. The formula is as follows:

$$\text{Att}^{(m)}(\mathbf{v}^{(l)}) = \text{Softmax} \left(\frac{(\mathbf{W}_Q^{(m)} \mathbf{v}^{(l)})(\mathbf{W}_K^{(m)} \mathbf{v}^{(l)})^\top}{\sqrt{d_k}} \right) \mathbf{W}_V^{(m)} \mathbf{v}^{(l)}, \quad (12)$$

where m represents the attention head index, $\mathbf{W}^{(m)}$ is the learnable projection matrices and d_k is the dimensionality of the features. Subsequently, the processed feature representations are normalized and subjected to inter-layer regularization. The formula is as follows:

$$\hat{\mathbf{v}}^{(l)} = \gamma^{(l)} \cdot \frac{\text{Att}^{(m)}(\mathbf{v}^{(l)})}{|\text{Att}^{(m)}(\mathbf{v}^{(l)})|_2} + \beta^{(l)}, \quad (13)$$

where $\gamma^{(l)}$ and $\beta^{(l)}$ are the learnable scaling and shifting parameters. The final multi-scale image representations are constructed through tensor concatenation, enabling feature reuse:

$$\mathbf{I}^l = \text{Concat}(\mathbf{v}^{(0)}, \hat{\mathbf{v}}^{(1)}, \hat{\mathbf{v}}^{(2)} \dots, \hat{\mathbf{v}}^{(L)}), \quad (14)$$

where $\mathbf{v}^{(0)}$ represents the input visual features $\mathbf{v}$, $\mathbf{I}^l$ represents the multi-scale visual embedding representation obtained from the image features through the above steps, $\text{Concat}(\cdot)$ refers

to concatenation along the channel dimension and L denotes the total number of layers for obtaining multi-scale features.

Based on the aforementioned method, the textual modal features T^f (generated fusion textual features) and T^c (entity-enriched textual features) are processed using the same multi-scale approach (by Eq. 11–14) as the visual features, resulting in the multi-scale generated fusion textual embeddings T^{fl} and the multi-scale entity-enriched textual embeddings T^{cl} . These two multi-scale textual representations preserve the global semantic structure of the original fused features and the local fine-grained associations after entity enrichment. Through a dual-path architecture, they achieve progressive alignment of visual-language modalities in a multi-level semantic space, ultimately forming complementary cross-modal representations.

E. Objective Function

To accurately measure cross-modal semantic similarity and enhance the discriminative ability of image-text matching, we propose an objective function that integrates multi-scale similarity metrics with triplet loss optimization. We first adopt a multi-scale weighted evaluation strategy: given multi-scale visual embedding v^l and multi-scale textual embedding, including multi-scale generated fusion textual embedding t^{fl} and multi-scale entity-enriched textual embedding t^{cl} , we introduce learnable weights $\lambda^{(l)}$ and attention matrices $W^{(l)} \in \mathbb{R}^{d^l \times d^l}$ at each multi-scale feature layer l . The normalized similarity is computed as follows:

$$\begin{aligned} Z(v^l, t^{fl}) &= \frac{1}{L} \sum_{l=0}^L \lambda^{(l)} \cdot (t^{fl(l)} W^{(l)} v^{l(l)T}), \\ Z(v^l, t^{cl}) &= \frac{1}{L} \sum_{l=0}^L \lambda^{(l)} \cdot (t^{cl(l)} W^{(l)} v^{l(l)T}). \end{aligned} \quad (15)$$

To minimize the distance between matched pairs while maximizing that of mismatched pairs, we construct a dual triplet loss [51], [52] function:

$$\begin{aligned} L_{triplet1} &= \left[\epsilon - Z(v^l, t^{fl}) + Z(v^l, t^{fl-}) \right]_+ \\ &\quad + \left[\epsilon - Z(t^{fl}, v^l) + Z(t^{fl}, v^{l-}) \right]_+, \\ L_{triplet2} &= \left[\epsilon - Z(v^l, t^{cl}) + Z(v^l, t^{cl-}) \right]_+ \\ &\quad + \left[\epsilon - Z(t^{cl}, v^l) + Z(t^{cl}, v^{l-}) \right]_+, \end{aligned} \quad (16)$$

where ϵ is the margin, $[\cdot]_+ = \max(\cdot, 0)$, t^{fl} and t^{cl} denote multi-scale generated fusion textual embeddings and multi-scale entity-enriched textual embeddings, respectively, and v^l represents multi-scale visual embeddings. The overall objective combines text semantic enhancement, dynamic feature fusion, and shared-space projection:

$$L_{total} = \mu L_{triplet1} + \eta L_{triplet2}, \quad (17)$$

where μ and η represent the weights balancing different loss functions. The full VPKE model is detailed in Algorithm 1.

Algorithm 1 Learning Algorithm for VPKE Model

Input: Training set: IMG_{train}, TXT_{train} ; hyperparameters: $\mu, \eta, \epsilon, m, G_k, L$; batch size $batch$, parameters.

Output: Trained VPKE Model.

Initialize parameters: θ_T via Eq. (1), θ_I via Eq. (1), Q via Eq. (3), w_m via Eq. (8), $W_C^{(l)}$ via Eq. (11), $\gamma^{(l)}, \beta^{(l)}$ via Eq. (13), and $\lambda^{(l)}, W^{(l)}$ via Eq. (15)

while not converged **do**

1. Obtain the initial features T and I for text and image modalities via Eq. (1).
2. Generate auxiliary caption features $\tilde{T}$ and key entity features C via Eq. (2)–Eq. (8).
3. Perform dynamic progressive fusion of multiple textual features to obtain T^f and T^c via Eq. (9)–Eq. (10).
4. Extract multi-scale features from image and text using convolutional layers and multi-head self-attention to obtain T^{fl}, T^{cl} , and I^l via Eq. (11)–Eq. (14).
5. Calculate the total loss L_{total} using the triplet losses in Eq. (17).
6. Update the network parameters $\theta_T, \theta_I, Q, w_m, W_C^{(l)}, \gamma^{(l)}, \beta^{(l)}, \lambda^{(l)}$, and $W^{(l)}$ by backpropagation.

end while

return The trained VPKE model

IV. EXPERIMENTS

To evaluate the effectiveness of the proposed method, image-text retrieval experiments are performed on the MSCOCO and Flickr30k datasets, with thorough comparisons to state-of-the-art approaches. Through experiments such as ablation studies, generalization tests, and visualization analysis, the effectiveness of the method is verified. Experiments are run on an NVIDIA L40 GPU using CLIP-based ViT-B/16 and ViT-L/14 architectures. The boundary threshold ϵ is set to 0.5, and m is set to 4. The learning rates are $1e-6$ for CLIP fine-tuning and $3e-5$ for the overall framework. The feature dimension is 512 and the similarity embedding dimension is 256. The batch size is 64 for training and 128 for testing. The model is trained for 20 epochs on both datasets.

A. Overview of Evaluation Metrics and Datasets

The evaluation employs the standard Recall at K (R@K) metric for image-text alignment tasks, measuring the proportion of correct matches among the top K retrieved results ($K=1, 5, 10$). Following prior work [53], [54], the rSum metric is additionally introduced. The experiments utilize the standard MSCOCO [55] and Flickr30K [56] benchmark datasets. MSCOCO contains 123K images with 5 captions each, split into 113K training, 5K validation and 5K test images. Flickr30K consists of 31K images with 5 captions per image, divided into 29K training, 1K validation and 1K test sets. The evaluation follows standard protocols and aligns with [57], reporting results on both the 1K test set (5-fold average) and full 5K test set of MSCOCO dataset.

B. Comparative Evaluation Methods

We evaluate the proposed method and compare it with current state-of-the-art image-text alignment approaches

TABLE II
PERFORMANCE ON MS-COCO 1K AND MSCOCO 5K. † INDICATES THE IMPROVED RESULTS
OBTAINED BY FINE-TUNING CLIP. * INDICATES ZERO-SHOT PERFORMANCE

Methods	MSCOCO (1k) dataset							MSCOCO (5k) dataset						
	Text Retrieval			Image Retrieval			rSum	Text Retrieval			Image Retrieval			rSum
	R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
<i>(Bert, partial pre-training)</i>														
MV-VSE _{IJCAI'22} [36]	80.4	96.6	99.0	64.9	91.2	96.0	528.1	59.1	86.3	92.5	42.5	72.8	83.1	436.3
VSRN++ _{TPAMI'22} [35]	77.9	96.0	98.5	64.1	91.0	96.1	523.6	54.7	82.9	90.9	42.0	72.2	82.7	425.4
CHAN _{CVPR'23} [8]	81.4	96.9	98.9	66.5	92.1	96.7	532.6	59.8	87.2	93.3	44.9	74.5	84.2	443.9
HREM _{CVPR'23} [26]	82.9	96.9	99.0	67.1	92.0	96.6	534.6	64.0	88.5	93.7	45.4	75.1	84.3	450.9
TVRN _{TMM'24} [37]	81.1	96.4	98.8	67.7	92.3	97.1	533.4	61.1	86.3	92.5	45.5	75.0	84.8	445.2
RANet _{TCSVT'24} [41]	82.8	97.5	98.9	69.6	93.2	97.2	539.2	60.6	86.3	93.0	45.3	75.6	85.5	446.3
JIIGNet _{TMM'24} [13]	82.1	98.4	99.1	67.0	93.8	98.1	538.5	58.1	86.8	91.2	41.9	75.2	84.7	437.9
RVSE++ _{TCSVT'25} [58]	81.6	96.6	98.8	66.6	92.1	96.6	532.4	60.6	86.4	92.8	44.5	74.5	84.5	443.4
VPKE (ours)	83.2	98.1	99.1	69.7	93.6	98.2	541.9	65.2	88.5	93.8	46.7	75.6	85.7	455.5
<i>(Dual-Encoder, pre-training)</i>														
CLIP* _{ICML'21} [15]	75.0	94.2	97.8	59.5	84.8	92.2	503.5	58.4	81.5	88.1	37.8	62.4	72.2	400.4
USER _{TIP'24} [6]	83.7	96.7	99.0	67.8	91.2	95.8	534.2	67.6	88.4	93.5	47.7	75.1	83.7	456.0
I-CLIP _{MM'23} [25]	79.8	96.9	99.1	66.7	89.6	95.2	527.3	63.0	85.5	91.4	47.3	72.7	81.2	441.1
ESL _{TCSVT'24} [10]	84.3	97.0	99.1	71.7	93.2	97.4	542.6	68.1	89.0	94.1	51.7	78.0	86.3	467.2
LG_MGC _{MM'24} [38]	-	-	-	-	-	-	-	67.6	88.5	93.8	51.2	77.9	86.4	465.3
LG _{TMM'24} [59]	80.7	96.0	98.5	67.1	91.1	96.2	529.6	62.1	85.6	92.1	47.1	73.9	83.2	443.9
DBL _{TIP'24} [60]	84.1	97.0	-	68.8	91.6	-	536.7	68.3	88.8	-	50.1	75.4	-	460.1
VPKE (ours)	85.3	97.5	99.1	72.9	93.2	97.6	545.6	68.9	89.2	94.1	52.6	78.3	86.5	469.7
<i>(Dual-Encoder, pre-training VLM)</i>														
CLIP [†] _{Vit-B/16}	83.6	96.8	99.0	71.1	93.0	97.2	540.7	66.9	88.3	93.6	51.5	78.0	86.1	464.4
+VPKE (ours)	85.5	97.6	99.1	73.0	93.1	97.7	546.0	69.2	89.0	94.2	52.8	78.5	86.5	470.2
CLIP [†] _{Vit-L/14}	86.5	98.0	99.4	74.9	94.6	98.0	551.4	68.6	89.9	94.7	53.9	80.0	87.9	475.1
+VPKE (ours)	87.1	98.2	99.3	75.9	95.0	98.0	553.5	73.6	91.8	95.9	58.1	82.6	89.4	491.4

addressing the issue from perspectives of information discrepancy and weak alignment, including MV-VSE [36], VSRN++ [35], CHAN [8], HREM [26], TVRN [37], RANet [41], JIIGNet [13], I-CLIP [25], USER [6], ESL [10], LG-MGC [38], LG [59], DBL [60] and the large-scale vision-language pre-training model CLIP [15]. The results for the comparison methods are obtained by executing the source code provided in the original papers or by referring to the experimental results reported therein. The best results are indicated in bold. Table. II and Table. III show a comparison of model performance with different feature encoding methods on the MS-COCO and Flickr30K datasets. These feature encoding methods are mainly divided into two categories: (i) Partial pre-trained models (BERT); (ii) Dual-encoder architecture pre-trained models (CLIP). The following is a brief description of these results: (1) On the Flickr30K test set, our method achieves significant improvements over all models. Compared with ESL, LG-MGC and I-CLIP methods, R@1 for text retrieval improves by 0.4%, 0.7% and 2.1% respectively, while for image retrieval, R@1 increases by 1.7%, 2.9% and 7.5% respectively. Additionally, CLIP_{Vit-B/16}+VPKE achieves the highest R@1, R@5, and R@10 scores of 93.7, 99.2, and 99.8, respectively, for text retrieval, and 82.0, 95.7, and 98.2 for image matching. CLIP_{Vit-L/14}+VPKE achieves the highest R@1, R@5, and R@10 scores of 95.4, 99.7, and 99.9 for text retrieval, and 84.8, 97.4, and 98.8 for image matching. Specifically, the rSum of CLIP_{Vit-B/16}+VPKE increases by 6.3%, and the rSum of CLIP_{Vit-L/14}+VPKE increases by 9.8%. CLIP_{Vit-B/16}+VPKE improves R@1 for text retrieval by 2.2% and for image retrieval by 2.7%. Similarly, CLIP_{Vit-L/14}+VPKE improves R@1 for text retrieval by 2.4% and for image

TABLE III
PERFORMANCE ON FLICKR30K. † INDICATES THE IMPROVED
RESULTS OBTAINED BY FINE-TUNING CLIP.
* INDICATES ZERO-SHOT PERFORMANCE

Methods	Text Retrieval			Image Retrieval			rSum
	R@1	R@5	R@10	R@1	R@5	R@10	
<i>(Bert, partial pre-training)</i>							
MV-VSE _{IJCAI'22} [35]	82.1	95.8	97.9	63.1	86.7	92.3	517.5
VSRN++ _{TPAMI'22} [35]	79.2	94.6	97.5	60.6	85.6	91.4	508.9
CHAN _{CVPR'23} [8]	80.6	96.1	97.8	63.9	87.5	92.6	518.5
HREM _{CVPR'23} [26]	84.0	96.1	98.6	64.4	88.0	93.1	524.2
TVRN _{TMM'24} [37]	82.1	95.6	98.3	63.9	87.6	92.6	520.1
RANet _{TCSVT'24} [41]	83.3	97.2	98.6	67.4	90.1	94.3	531.0
JIIGNet _{TMM'24} [13]	83.0	96.8	98.9	62.8	89.0	92.0	522.5
RVSE++ _{TCSVT'25} [58]	83.6	96.5	98.6	64.3	88.2	93.0	524.2
VPKE (ours)	84.3	97.5	99.1	68.3	90.5	93.6	533.3
<i>(Dual-Encoder, pre-training)</i>							
CLIP [*] _{ICML'21} [15]	88.0	98.7	99.4	68.7	90.6	95.2	540.6
USER _{TIP'24} [6]	86.3	97.6	99.4	69.5	91.0	94.4	538.1
I-CLIP _{MM'23} [25]	91.2	99.2	99.7	74.3	92.3	95.5	552.2
ESL _{TCSVT'24} [10]	92.9	99.2	99.7	80.1	95.4	97.9	565.2
LG_MGC _{MM'24} [38]	92.6	99.5	99.7	78.9	95.5	98.2	564.5
LG _{TMM'24} [59]	87.9	98.8	99.7	75.9	94.4	96.9	553.5
DBL _{TIP'24} [60]	92.8	99.5	-	78.9	95.2	-	563.3
VPKE (ours)	93.3	99.5	99.9	81.8	95.9	98.3	568.7
<i>(Dual-Encoder, pre-training VLM)</i>							
CLIP [†] _{Vit-B/16}	91.5	99.2	99.5	79.3	95.2	97.6	562.3
+VPKE (ours)	93.7	99.2	99.8	82.0	95.7	98.2	568.6
CLIP [†] _{Vit-L/14}	93.0	99.2	99.8	80.4	95.7	98.1	566.2
+VPKE (ours)	95.4	99.7	99.9	84.8	97.4	98.8	576.0

retrieval by 4.4%. (2) On the MSCOCO 5K test set, compared with ESL, LG-MGC and I-CLIP methods, R@1 for text retrieval increases by 0.8%, 1.3%, and 5.9% respectively, while for image retrieval, R@1 rises by 0.9%, 1.4%, and 5.3% respectively. Additionally, CLIP_{Vit-B/16}+VPKE achieves the highest R@1, R@5, and R@10 scores of 69.2, 89.0, and 94.2 for text retrieval, respectively, and 52.8, 78.5, and 86.5 for image matching. CLIP_{Vit-L/14}+VPKE achieves the highest

TABLE IV
ABLATION EXPERIMENT RESULTS ON FLICKR30K.
BASELINE MODEL IS CLIP_{VIT-L/14}[†]

Methods	IG	AR	VR	MHA	Text Retrieval		Image Retrieval	
					R@1	R@5	R@1	R@5
Baseline					93.0	99.2	80.4	95.7
(Effectiveness of MHA)								
with IG	✓				92.5	99.2	79.6	95.2
w/o AR&VR	✓			✓	93.0	99.1	81.0	95.8
w/o MHA	✓	✓	✓		94.4	99.4	83.9	97.1
(Effectiveness of VR)								
w/o VR	✓	✓		✓	94.9	99.8	84.1	97.1
(Effectiveness of CKRE)								
w/o VR&MHA	✓	✓			94.1	99.4	82.5	96.0
w/o AR	✓		✓	✓	94.8	99.7	84.3	97.2
VPKE	✓	✓	✓	✓	95.4	99.7	84.8	97.4

R@1, R@5, and R@10 scores of 73.6, 91.8, and 95.9 for text retrieval, and 58.1, 82.6, and 89.4 for image matching. Specifically, the rSum of CLIP_{Vit-B/16}+VPKE increases by 5.8%, and the rSum of CLIP_{Vit-L/14}+VPKE increases by 16.3%. CLIP_{Vit-B/16}+VPKE improves R@1 for text retrieval by 2.3% and for image retrieval by 1.3%. CLIP_{Vit-L/14}+VPKE improves R@1 for text retrieval by 5.0% and for image retrieval by 4.2%. Similar improvements appear on the MSCOCO 1K dataset. Experiments demonstrate that the VPKE method significantly outperforms the original pre-trained model and mainstream cross-modal heterogeneity bridging methods. It effectively reduces the difference between cross-modal entities through a visual perception framework, enhancing retrieval performance.

C. Ablation Analysis and Heterogeneity Evaluation

1) *Model Ablation Experiments*: To validate the effectiveness of the three modules in the VPKE framework, as shown in Table. IV, we conduct extensive ablation experiments. For simplicity, we denote CKRE, VR, and MHA as the three modules in the VPKE framework: “Cross-Modal Knowledge Retrieval Enhancement Network,” “Vision-Guided Semantic Rebalancing Module,” and “Multi-scale Heterogeneous Attention Aggregator,” respectively. CKRE consists of two components: the image caption generator (IG) and the adaptive knowledge refinement module (AR). As shown in Table. IV, we investigate the design of each module in subsections I) - 3). For simplicity, the experiments are based on a CLIP-based dual encoder, with the fine-tuned CLIP_{Vit-L/14} serving as the baseline.

a) *Effectiveness of CKRE*: CKRE consists of two parts: the image caption generator (IG) and the adaptive knowledge refinement module (AR). (1) AR plays a crucial role in the ablation experiments. When this module is removed (w/o AR), the performance of model drops significantly (R@1 for text and image retrieval dropped by 0.6% and 0.5%, respectively). The reason lies in the AR module integrating RAG and key entity extraction, which filters noise and collaboratively optimizes key cross-modal features when external knowledge is introduced, thereby ensuring knowledge relevance and accuracy. Without the AR module, the generated text may fail to align well with the image content, leading to a decrease in descriptive accuracy. By incorporating visual features, the AR

module can more accurately guide text generation, thereby significantly improving performance in retrieval tasks. (2) IG is used to generate auxiliary captions of images. Experimental results show that when IG is used alone (with IG), the performance of model significantly decreases (R@1 for image retrieval dropped by 5.2%). This is because the generated captions may contain abstract concepts or objects unrelated to the task, or exhibit slight semantic redundancy with the original text. These factors increase noise and impact overall performance. In particular, vague or inaccurate captions hinder accurate understanding of the image content, leading to semantic inconsistency between the text and image. Therefore, the close collaboration between the IG and the AR (Compared to with IG, the text retrieval R@1 for w/o VR&MHA improved by 1.6%) ensures that the generated content maintains high semantic consistency with the image content, thereby enhancing retrieval performance.

b) *Effectiveness of VR*: The experimental results show that when this module is absent (w/o VR), the performance of the VPKE model decreases (R@1 for text and image retrieval dropped by 0.5% and 0.7%, respectively). This is because the VE module dynamically adjusts the contribution weights of the original texts, key entities, and auxiliary captions through multi-level similarity evaluation. This method enables the adaptive fusion of different semantic features, enhancing the flexibility and adaptability of the model. The absence of the VR module weakens the integration of image and text features, leading to inadequate cross-modal interaction understanding and consequently impairing the accuracy of image-text matching.

c) *Effectiveness of MHA*: The MHA module plays a key role in improving the model understanding of cross-modal relationships. Without this module (w/o MHA), the model fails to effectively integrate semantic information from multiple levels, resulting in poor performance when dealing with complex cross-modal tasks (R@1 for text and image retrieval dropped by 1.0% and 0.9%, respectively). The MHA module performs multi-scale feature extraction of semantic information from multiple levels, enhancing the ability of cross-modal semantic representation. It enables the model to fully capture the deep relationships between images and text, enhancing the overall semantic representation, particularly when handling complex cross-modal relationships. Moreover, the absence of AR&VR (w/o AR&VR) and the use of IG module again validate the effectiveness of MHA module (The R@1 for text retrieval improved by 0.4%). Even when the model performance significantly decreases due to the use of IG alone (with IG), the MHA module still demonstrates strong performance.

2) *Modality Heterogeneity Analysis*: As shown in Fig. 5, to verify the effectiveness of the proposed method, fine-tuning is performed on the Flickr30K dataset. A random selection of 1000 image-text pairs from its test set is used to generate image and text features, followed [12] by dimensionality reduction analysis using TSNE. Comparative experiments with the CLIP_{Vit-B/16}[†] model demonstrate that the features generated by our method (VPKE) exhibit a significant distribution difference from the original features. This approach

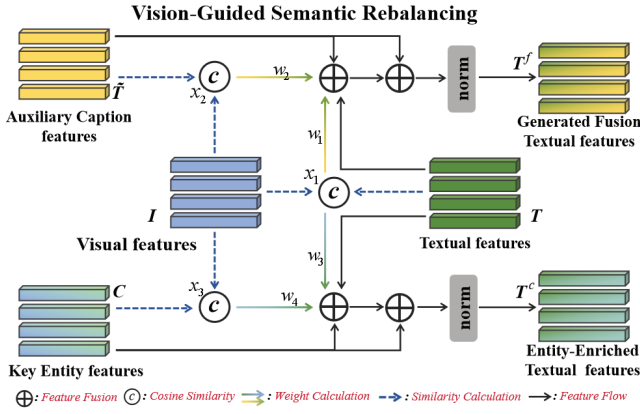


Fig. 4. Vision-guided semantic rebalancing strategy. Through multi-level similarity evaluation, the contribution weights of the textual features, key entity features, and auxiliary caption features are adjusted dynamically and progressively.

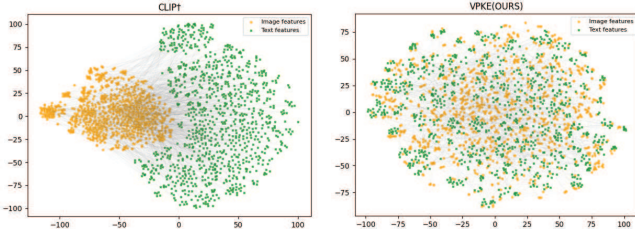


Fig. 5. Visualization of modality heterogeneity. Orange represents the visualization of image features, while green denotes the visualization of text features. The visualization displays features generated from 1000 randomly selected image-text pairs in the Flickr30K test set.

not only modifies the structure of the feature space but also efficiently clusters similar samples while distinguishing dissimilar ones, which significantly improves the ability of the model to recognize similar samples. Furthermore, in the modality alignment analysis, we observe that the baseline method clearly separates text (green) and image (orange) features. In contrast, our method incorporates auxiliary visual knowledge, leading to tightly intermingled feature points from both modalities in the embedding space. This results in a significantly expanded overlapping region between the two modalities. The semantic representation capability of text features is effectively enhanced, thereby significantly reducing modality heterogeneity. The experiments collectively confirm the superiority of the proposed method in improving image-text retrieval performance from the perspectives of feature distribution optimization and modality alignment enhancement.

D. Efficiency Comparison and Generalization Analysis

1) *Efficiency Analysis*: (1) To conduct a comprehensive comparison, we evaluate the matching performance of representative models on the Flickr30K test set. All models are tested under identical settings, including the same feature encoder and experimental environment, with a batch size of 1000. The VPKE model was compared with advanced image-text retrieval models, including CLIP [15], I-CLIP [25],

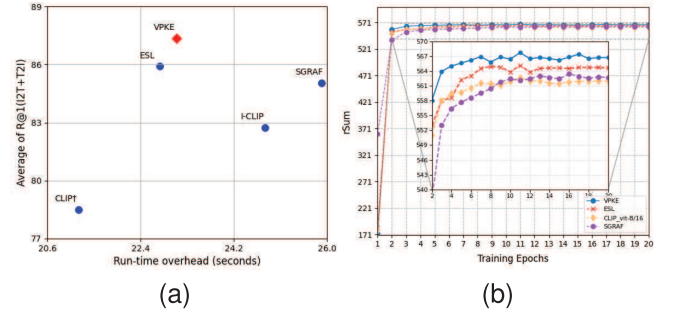


Fig. 6. The matching performance and speed comparison on the Flickr30K test set were conducted using an NVIDIA L40 GPU for all methods (based on the CLIP Dual-Encoder). (a) Retrieval efficiency (b) Model convergence analysis.

TABLE V

COMPLEXITY COMPARISON ON THE FLICKR30K TEST SET. PARAMETER COUNT (PARAMS). TO FACILITATE A MORE DETAILED COMPARISON, RESULTS EXCLUDING THE ENCODING MODULE ARE PROVIDED, DENOTED AS PARAMS W/O ENC

	SGRAF	ESL	I-CLIP	VPKE
Params (M)	150.8	150.3	151.0	149.9
Params w/o enc (M)	1.25	0.69	1.03	0.34

TABLE VI

EVALUATION OF TEXT SEMANTIC DIVERSITY IS CONDUCTED ON THE MSCOCO AND FLICKR30K DATASETS

	Datasets	Original texts	Auxiliary captions	Integrated captions (VPKE)
ECE	flickr30k	1.58	1.52	3.25
	mscoco	1.61	1.57	3.31
SDE	flickr30k	0.9414	0.9405	0.9553 (1.5%↑ / 1.6%↑)
	mscoco	0.9438	0.9431	0.9559 (1.3%↑ / 1.4%↑)

ESL [10], and SGRAF [61]. The matching speed of all models was reported based on re-implemented open-source code under uniform settings. As shown in Fig. 6 (a), VPKE outperforms the contrastive models in both speed and performance. Although ESL has a slightly shorter runtime, VPKE demonstrates better accuracy. VPKE not only excels in effectiveness but also performs exceptionally in efficiency. The convergence analysis in Fig. 6 (b) demonstrates that VPKE achieves rapid performance improvement during early training stages while maintaining stable convergence, indicating efficient learning capability. In comparison, the baseline methods show slow performance improvement and significant fluctuations, highlighting the robustness and optimization capability of VPKE when handling complex image-text retrieval tasks. (2) A complexity comparison is shown in Table. V. To ensure fairness, all three methods use the same visual and textual feature extraction modules. The total model parameters (Params) and the parameters after removing the visual and textual encoding modules (Params w/o enc) are displayed. VPKE has fewer total parameters than other methods, and it also has fewer parameters after the encoding modules are removed. This indicates that VPKE maintains low complexity while still providing high performance.

2) *Generalization Performance*: To assess the generalization ability of the VPKE framework, we perform cross-dataset testing, as presented in Table. VII. This

TABLE VII

CROSS-DATASET EVALUATION FOR GENERALIZATION CAPABILITY
ABLATION STUDIES ABOUT OUR VPKE FRAMEWORK

Methods	Datasets		Text Retrieval			Image Retrieval		
	train	test	R@1	R@5	R@10	R@1	R@5	R@10
VPKE _{+c}	flickr30k	mscoco	82.3	96.4	98.8	69.1	91.5	96.4
VPKE _{+f}	mscoco	flickr30k	92.3	99.5	99.6	83.0	96.4	98.1

TABLE VIII

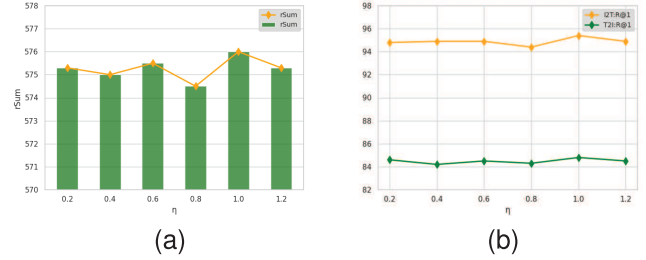
COMPARISON OF NOISE ROBUSTNESS BETWEEN VPKE AND CLIP ON THE
FLICKR30K TEST SET (NOISE RATIO IS QUANTIFIED BY
THE PROPORTION OF NOISE WORDS IN TEXT CAPTION)

Methods	Noise ratio test (1k)	Text Retrieval		Image Retrieval	
		R@1	R@5	R@1	R@5
CLIP _{+n0}	0%	85.5	98.2	72.8	93.1
VPKE _{+n0}	0%	89.1	98.4	78.0	94.7
CLIP _{+n1}	20%	81.3	97.0	72.4	92.8
VPKE _{+n1}	20%	88.6	98.1	77.6	94.6
CLIP _{+n2}	40%	76.6	96.1	71.6	92.2
VPKE _{+n2}	40%	88.2	98.6	77.3	94.3
CLIP _{+n3}	60%	75.5	96.1	71.3	92.1
VPKE _{+n3}	60%	88.1	98.1	77.0x	94.2
CLIP _{+n4}	80%	72.7	95.3	70.6	91.9
VPKE _{+n4}	80%	87.3	98.2	76.3	93.7

involves training the model on one dataset and evaluating it on a completely different dataset. This method highlights the robustness of the framework in transferring learned representations across datasets while achieving high performance in both text and image retrieval tasks. The consistent results across different datasets underscore the strong generalization ability of the framework, indicating its potential for broader applicability in visual and textual retrieval scenarios.

3) *Analysis of Noise Data Robustness:* As shown in Table. VIII, the image-text retrieval task is evaluated on the Flickr30k test set under different noise ratios (i.e., half of the text captions in the test set are noise samples, and the noise proportion is quantified by the proportion of noise words in each text caption). The experiment is conducted with 1,000 image-text sample pairs under the same experimental settings. We find that the VPKE method demonstrates stronger robustness against noise interference compared to CLIP [15]. As the noise level in captions rises, the retrieval performance of both methods decreases. However, VPKE shows a much slower rate of performance degradation. In the text retrieval task, VPKE consistently outperforms CLIP in retrieval accuracy. Its advantage is especially noticeable in high-noise environments, highlighting its better robustness to noisy textual captions. VPKE also maintains a stable lead in the image retrieval task, especially under significant noise conditions. Even in such challenging environments, it maintains a high retrieval success rate. The results show that VPKE effectively filters out noisy caption interference due to its enhanced noise modeling mechanism. This significantly improves retrieval reliability in noisy data conditions.

4) *Analysis of Textual Semantic Diversity:* To thoroughly validate the effectiveness of text semantic fusion for diversity enhancement under the VPKE framework in image-text retrieval, we evaluate performance through entity coverage

Fig. 7. The values of $rSum$ and $R@1$ varying with η on Flickr30K dataset.

and semantic space distribution. As shown in Table. VI, blue text denotes comparisons with original texts while red text indicates auxiliary captions comparisons. Experiments are conducted on 1,000 sample pairs using identical configurations. It reveals the advantages of the VPKE in improving the quality of retrieved texts through quantitative indicators. The Entity Coverage Entropy (ECE), based on information entropy theory, measures the diversity and distribution uniformity of entities in generated captions. The indicator performs better when the entity distribution is more extensive and balanced. This means it should cover diverse semantic entities without repetition or dominance by any single entity. Experiments show that VPKE significantly outperforms pure original texts and auxiliary captions, significantly reducing the phenomenon of single-entity dominance and improving the comprehensiveness of retrieval results. The Semantic Distribution Entropy (SDE) evaluates semantic richness by quantifying the distribution characteristics of text semantic vectors in high-dimensional space. If the text is evenly distributed in multiple semantic dimensions, the entropy value is higher, indicating richer semantics and more diverse expressions. In experiments, VPKE has achieved significant improvements (1.5% and 1.6%) in captions of complex scene images compared with traditional baselines (such as original texts and auxiliary captions), reflecting its advantages in expanding the boundaries of semantic expression and enhancing semantic richness. The cross-validation of the two-dimensional entropy indicators reveals the significant gain of VPKE on text diversity from both entity and semantic levels, further verifying the effectiveness of the method.

E. Visualization and Parameter Analysis

1) *Impact of Various Parameter Settings:* Under the proposed model framework, four adjustable hyperparameters are defined: μ , η , G_k , and L . To evaluate the impact of these parameters on retrieval performance, we plot the $rSum$ histograms and the $R@1$ curves. $rSum$ is a widely used evaluation metric, representing the sum of $R@1$, $R@5$, and $R@10$ in both image and text retrieval tasks. A higher $rSum$ value indicates better model performance. $R@1$ reflects the ability of the model to correctly identify the top match in a single retrieval attempt. A higher $R@1$ suggests better accuracy in retrieving the correct top result. Fig. 7 and Fig. 8 present the results of the loss function weight parameters on the Flickr30K dataset. From the histogram of μ in panel (a), it is observed

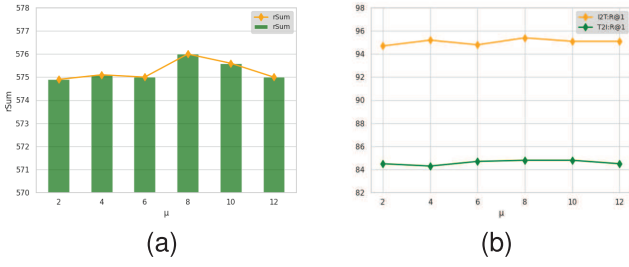
Fig. 8. The values of rSum and R@1 varying with μ on Flickr30K dataset.

TABLE IX
IMPACT OF G_k AND L PARAMETERS ON FLICKR30K. PARAMETER ANALYSIS USING CLIP VIT-L/14 ARCHITECTURE

Methods	Text Retrieval		Image Retrieval		rSum
	R@1	R@5	R@1	R@5	
Number of selected Top-K values					
$G_k = 2$	94.2	99.5	84.2	97.3	573.9
$G_k = 4$	94.3	99.4	84.4	97.2	574.1
$G_k = 6$	94.8	99.6	84.4	97.3	575.0
$G_k = 8$	95.6	99.7	84.1	97.3	575.4
$G_k = 10$	95.4	99.7	84.8	97.4	576.0
Number of selected L values					
$L = 2$	94.6	99.7	84.7	97.5	575.2
$L = 4$	94.9	99.7	84.3	97.4	575.0
$L = 6$	95.4	99.7	84.8	97.4	576.0
$L = 8$	95.2	99.7	84.8	97.2	575.6

that when μ is set to 8, VPKE achieves the best rSum performance, and also attains the highest R@1 value in panel (b). The curve for η indicates that retrieval performance is significantly influenced by the value of η , with optimal performance for both rSum and R@1 observed when η is set to 1.

The parameter G_k represents the number of concepts selected from the concept memory bank M_D . As shown in Table IX, when the value of G_k is small, the model tends to focus primarily on the most relevant concepts. This can result in an overemphasis on certain prominent concepts, while potentially overlooking less relevant information. By conducting experiments with values of G_k , it can be observed that as G_k increases, both the R@1 and R@5 metrics for text and image retrieval. When $G_k = 10$, the R@1 for text retrieval is 95.4, slightly lower than the R@1 of 95.6 when $G_k = 8$, but the R@1 for image retrieval increases to 84.8 with $G_k = 10$. The overall rSum value indicates that $G_k = 10$ (576.0) shows a greater advantage compared to $G_k = 8$ (575.4). This suggests that even with a slight decrease in certain metrics, such as R@1, $G_k = 10$ still outperforms $G_k = 8$ in overall performance. This is because $G_k = 10$ captures more comprehensive information, which enhances the overall retrieval capability. Therefore, assigning G_k a value of 10 is optimal.

The parameter L represents the number of layers used in the multi-scale feature extraction process. As L grows, the effectiveness of feature extraction improves significantly, particularly in text and image retrieval tasks. Specifically, when L is increased from 2 to 6, both the R@1 and R@5 metrics show consistent improvement. At $L = 6$,

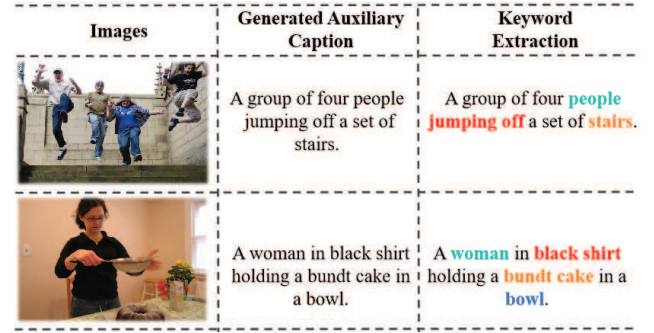


Fig. 9. Visualization of irrelevant words in the auxiliary captions generated by BLIP-2. Different colors (excluding black) are used to annotate the words or phrases extracted from the generated captions, which will be used to construct the external knowledge base.

R@1 reaches 95.4, indicating that the model is more effective at capturing and expressing the semantic relationships between images and texts at this level. However, when the value of L increases to 8, efficiency slightly decreases. This suggests that using too many layers may cause feature repetition or overfitting. Therefore, it is optimal to maintain L at 6.

2) *Visualized Verification of Adaptive Knowledge Refinement*: The effectiveness of the VPKE method is validated through multi-stage visualization analysis. As shown in Fig. 9, during the knowledge base construction phase, the auxiliary captions generated by the BLIP-2 model (e.g., “A group of four people jumping off a set of stairs.”) contain some irrelevant terms. We apply part-of-speech tagging and semantic analysis techniques to filter out noise semantic units and extract effective keywords or phrases (e.g., “jumping off,” “stairs”). These filtered semantic units are used to build a structured external knowledge base. In the semantic enhancement phase (as shown in Fig. 10), the system dynamically selects the top-k most relevant semantic units from the constructed knowledge base based on similarity confidence scores. For instance, for a building scene, words such as “large construction crane” and “zip line” are extracted (the visual correlation with image regions is established through color coding). Experimental results show that this method effectively expands the coverage and semantic density of the description while preserving the core meaning of the original texts.

3) *Visualization of Retrieval Results*: Fig. 11 shows qualitative results of $\text{CLIP}_{\text{vit-L/14}}^{\dagger}$ and the proposed VPKE model on text-to-image and image-to-text retrieval tasks. The upper part presents examples of text retrieval using images. Green sentences indicate matching examples, while red sentences represent non-matching examples. The lower part demonstrates examples of image retrieval using text. Green boxes highlight matching examples, while red boxes indicate non-matching ones. For the text-to-image retrieval task, we use the query text “A group of people playing on a beach.” to search for relevant images. From the retrieved images, it is clear that images with similar content are distinguishable because the adaptive knowledge refinement module allows the text side to learn semantic information from the image, leading

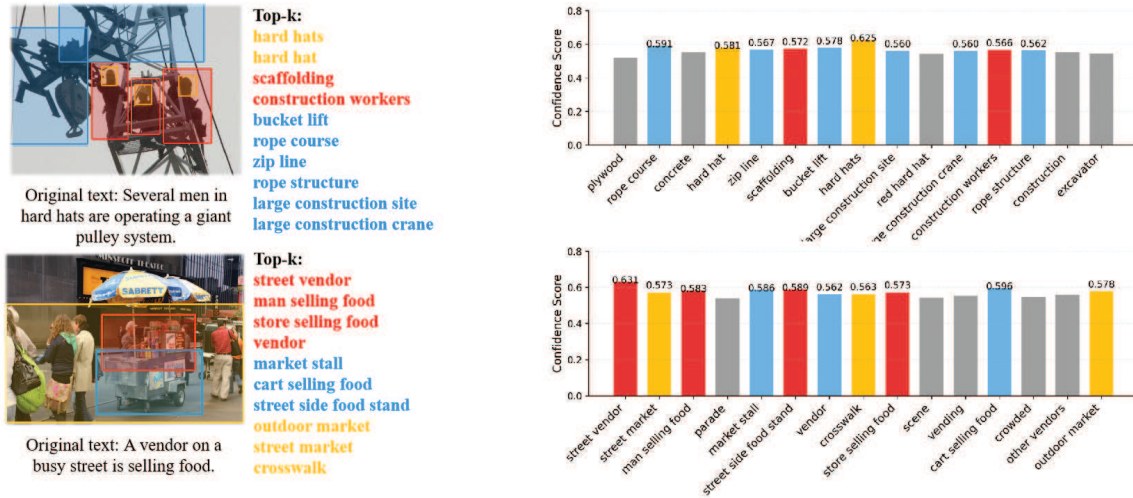


Fig. 10. Visualization of dynamic selection in the semantic enhancement unit of the adaptive knowledge refinement module. Dual-confidence evaluation based on similarity, the top-k most relevant semantic units are selected from the concept memory bank M_D .



Fig. 11. Qualitative retrieval results for sample queries from the Flickr30K dataset. In this figure, images retrieval by text shows the top 4 retrieval results, and texts retrieval by image shows the top 3 results. Green texts and images represent matches with the ground truth, while red indicates false positives.

to more accurate matching. Similarly, for the image-to-text retrieval task, our model retrieves the correct text with high accuracy.

V. CONCLUSION

In this paper, we propose a vision-perception knowledge enhancement framework to address multimodal information heterogeneity. Leveraging BLIP-2, the framework extracts latent image semantics and generates auxiliary captions, overcoming traditional text-modality bottlenecks. We introduce a novel adaptive knowledge refinement mechanism combining RAG with key entity extraction, filtering noise while optimizing cross-modal features for relevance and accuracy. Multi-level similarity assessment dynamically adjusts fusion weights between original texts, key entities, and auxiliary captions, enabling adaptive semantic fusion. Additionally, multi-scale feature extraction strengthens cross-modal representation. Future work will explore applications in zero-shot learning and open-domain tasks, further enhancing multimodal feature integration via vision-language models to improve efficiency.

REFERENCES

- [1] H. Sun, Y. Cao, and G. Liu, "Self-supervised cluster-contrast distillation hashing network for cross-modal retrieval," *IEEE Access*, vol. 11, pp. 96584–96593, 2023.
- [2] M. Cheng et al., "ViSTA: Vision and scene text aggregation for cross-modal retrieval," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5174–5183.
- [3] J. I. Wattasseril, S. Shekhar, J. Döllner, and M. Trapp, "Zero-shot video moment retrieval using BLIP-based models," in *Proc. Int. Symp. Vis. Comput. (ISVC)*, 2023, pp. 160–171.
- [4] K. Zhang, F. Wu, G. Zhang, J. Liu, and M. Li, "BVA-transformer: Image-text multimodal classification and dialogue model architecture based on blip and visual attention mechanism," *Displays*, vol. 83, Jul. 2024, Art. no. 102710.
- [5] J. Zhuang, J. Yu, Y. Ding, X. Qu, and Y. Hu, "Towards fast and accurate image-text retrieval with self-supervised fine-grained alignment," *IEEE Trans. Multimedia*, vol. 26, pp. 1361–1372, 2024.
- [6] Y. Zhang, Z. Ji, D. Wang, Y. Pang, and X. Li, "USER: Unified semantic enhancement with momentum contrast for image-text retrieval," *IEEE Trans. Image Process.*, vol. 33, pp. 595–609, 2024.
- [7] D. Feng, X. He, and Y. Peng, "MKVSE: Multimodal knowledge enhanced visual-semantic embedding for image-text retrieval," *ACM Trans. Multimedia Comput., Commun., Appl.*, vol. 19, no. 5, pp. 1–21, Sep. 2022.
- [8] Z. Pan, F. Wu, and B. Zhang, "Fine-grained image-text matching by cross-modal hard aligning network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 19275–19284.
- [9] W. Wu, H. Luo, B. Fang, J. Wang, and W. Ouyang, "Cap4 video: What can auxiliary captions do for text-video retrieval," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 10704–10713.
- [10] K. Zhang, B. Hu, H. Zhang, Z. Li, and Z. Mao, "Enhanced semantic similarity learning framework for image-text matching," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 4, pp. 2973–2988, Apr. 2024.
- [11] H. Lu, N. Fei, Y. Huo, Y. Gao, Z. Lu, and J.-R. Wen, "COTS: Collaborative two-stream vision-language pre-training model for cross-modal retrieval," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 15671–15680.
- [12] H. Huang, Z. Nie, Z. Wang, and Z. Shang, "Cross-modal and uni-modal soft-label alignment for image-text retrieval," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2024, vol. 38, no. 16, pp. 18298–18306.
- [13] R. Pan, H. Yang, C. Li, and J. Yang, "Joint intra & inter-grained reasoning: A new look into semantic consistency of image-text retrieval," *IEEE Trans. Multimedia*, vol. 26, pp. 4912–4925, 2024.
- [14] H. Zhu, C. Zhang, Y. Wei, S. Huang, and Y. Zhao, "ESA: External space attention aggregation for image-text retrieval," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 10, pp. 6131–6143, Oct. 2023.

- [15] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.
- [16] J. Li, D. Li, C. Xiong, and S. Hoi, "BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 12888–12900.
- [17] J. Wang, P. Zhou, M. Z. Shou, and S. Yan, "Position-guided text prompt for vision-language pre-training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 23242–23251.
- [18] X. Xu, C. Wu, S. Rosenman, V. Lal, W. Che, and N. Duan, "BridgeTower: Building bridges between encoders in vision-language representation learning," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2023, vol. 37, no. 9, pp. 10637–10647.
- [19] M. Cao, Y. Bai, Z. Cao, L. Nie, and M. Zhang, "Efficient image-text retrieval via keyword-guided pre-screening," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 6, pp. 5132–5145, Jun. 2024.
- [20] J. Yi, Y. Zhu, J. Xie, and Z. Chen, "Cross-modal variational auto-encoder for content-based micro-video background music recommendation," *IEEE Trans. Multimedia*, vol. 25, pp. 515–528, 2023.
- [21] D. Wu et al., "Language prompt for autonomous driving," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2025, vol. 39, no. 8, pp. 8359–8367.
- [22] Z. Yang, L. Chen, Y. Sun, and H. Li, "Visual point cloud forecasting enables scalable autonomous driving," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 14673–14684.
- [23] J. Li, D. Li, S. Savarese, and S. C. H. Hoi, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2023, pp. 19730–19742.
- [24] Y. Gao et al., "Retrieval-augmented generation for large language models: A survey," 2023, *arXiv:2312.10997*.
- [25] X. Dong, L. Han, D. Zhang, L. Liu, J. Han, and H. Zhang, "Giving text more imagination space for image-text matching," in *Proc. 31st ACM Int. Conf. Multimedia*, Oct. 2023, pp. 6359–6368.
- [26] Z. Fu, Z. Mao, Y. Song, and Y. Zhang, "Learning semantic relationship among instances for image-text matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 15159–15168.
- [27] X. Wei, T. Zhang, Y. Li, Y. Zhang, and F. Wu, "Multi-modality cross attention network for image and sentence matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10938–10947.
- [28] Q. Zhang, Z. Lei, Z. Zhang, and S. Z. Li, "Context-aware attention network for image-text retrieval," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3533–3542.
- [29] J. Wei, X. Xu, Y. Yang, Y. Ji, Z. Wang, and H. T. Shen, "Universal weighting metric learning for cross-modal matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 13005–13014.
- [30] F. Faghri, D. J. Fleet, J. R. Kiros, and S. Fidler, "VSE++: Improving visual-semantic embeddings with hard negatives," 2017, *arXiv:1707.05612*.
- [31] Z. Liu, F. Chen, J. Xu, W. Pei, and G. Lu, "Image-text retrieval with cross-modal semantic importance consistency," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 5, pp. 2465–2476, May 2023.
- [32] K. Zhang, L. Zhang, B. Hu, M. Zhu, and Z. Mao, "Unlocking the power of cross-dimensional semantic dependency for image-text matching," in *Proc. 31st ACM Int. Conf. Multimedia*, Oct. 2023, pp. 4828–4837.
- [33] X. Zhang, X. Niu, P. Fournier-Viger, and X. Dai, "Image-text retrieval via preserving main semantics of vision," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2023, pp. 1967–1972.
- [34] H. Xu, J. Liao, H. Liu, and Y. Sun, "Learning semantic alignment using global features and multi-scale confidence," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 2, pp. 897–910, Feb. 2024.
- [35] K. Li, Y. Zhang, K. Li, Y. Li, and Y. Fu, "Image-text embedding learning via visual and textual semantic reasoning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 641–656, Jan. 2023.
- [36] Z. Li, C. Guo, Z. Feng, J.-N. Hwang, and X. Xue, "Multi-view visual semantic embedding," in *Proc. 31st Int. Joint Conf. Artif. Intell.*, Jul. 2022, pp. 1130–1136.
- [37] S. Pang, Y. Zeng, J. Zhao, and J. Xue, "A mutually textual and visual refinement network for image-text matching," *IEEE Trans. Multimedia*, vol. 26, pp. 7555–7566, 2024.
- [38] F. Zhang, S. Qu, F. Shi, and C. Xu, "Overcoming the pitfalls of vision-language model for image-text retrieval," in *Proc. 32nd ACM Int. Conf. Multimedia*. New York, NY, USA: Association for Computing Machinery, Oct. 2024, pp. 2350–2359.
- [39] H. Zhang, L. Zhang, K. Zhang, and Z. Mao, "Identification of necessary semantic undertakers in the causal view for image-text matching," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2024, vol. 38, no. 7, pp. 7105–7114.
- [40] Z. Li, L. Zhang, K. Zhang, Y. Zhang, and Z. Mao, "Fast, accurate, and lightweight memory-enhanced embedding learning framework for image-text retrieval," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 7, pp. 6542–6558, Jul. 2024.
- [41] G. Xiong, M. Meng, T. Zhang, D. Zhang, and Y. Zhang, "Reference-aware adaptive network for image-text matching," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 10, pp. 9678–9691, Oct. 2024.
- [42] W. Zhang, X. Zhao, L. Zhao, D. Yin, G. H. Yang, and A. Beutel, "Deep reinforcement learning for information retrieval: Fundamentals and advances," in *Proc. 43rd Int. ACM SIGIR Conf. Res. Develop. Inf. Retr.*, Jul. 2020, pp. 2468–2471.
- [43] K. Pham, C. Huynh, S.-N. Lim, and A. Shrivastava, "Composing object relations and attributes for image-text matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 14354–14363.
- [44] W. Han, G. Niu, M. Zhou, and X. Zhang, "Hierarchical bi-directional conceptual interaction for text-video retrieval," *Multimedia Syst.*, vol. 30, no. 6, p. 317, Dec. 2024.
- [45] H. Kang, X. Zhang, W. Han, and M. Zhou, "Dark knowledge association guided hashing for unsupervised cross-modal retrieval," *Multimedia Syst.*, vol. 30, no. 6, pp. 1–13, Dec. 2024.
- [46] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, "Language models are unsupervised multitask learners," *OpenAI Blog*, vol. 1, no. 8, p. 9, 2019.
- [47] X. Pu et al., "MiC: Image-text matching in circles with cross-modal generative knowledge enhancement," *Knowledge-Based Syst.*, vol. 289, Apr. 2024, Art. no. 111503.
- [48] M. Tschannen, B. Mustafa, and N. Houlsby, "CLIPPO: Image-and-language understanding from pixels only," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 11006–11017.
- [49] G. Cai et al., "Ask&confirm: Active detail enriching for cross-modal retrieval with partial query," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 1815–1824.
- [50] X. Li et al., "OSCAR: Object-semantics aligned pre-training for vision-language tasks," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 121–137.
- [51] C. Deng, Z. Chen, X. Liu, X. Gao, and D. Tao, "Triplet-based deep hashing network for cross-modal retrieval," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 3893–3903, Aug. 2018.
- [52] Y. Peng and J. Chi, "Unsupervised cross-media retrieval using domain adaptation with scene graph," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 11, pp. 4368–4379, Nov. 2020.
- [53] J. Chen, H. Hu, H. Wu, Y. Jiang, and C. Wang, "Learning the best pooling strategy for visual semantic embedding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15789–15798.
- [54] K.-H. Lee, X. Chen, G. Hua, H. Hu, and X. He, "Stacked cross attention for image-text matching," in *Proc. Eur. Conf. Comp. Vis.*, 2018, pp. 201–216.
- [55] T. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 740–755.
- [56] A. Karpathy, A. Joulin, and F. F. F. Li, "Deep fragment embeddings for bidirectional image sentence mapping," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2014, pp. 1889–1897.
- [57] A. Karpathy and L. Fei-Fei, "Deep visual-semantic alignments for generating image descriptions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3128–3137.
- [58] Z. Li, C. Guo, X. Wang, Z. Feng, and Z. Du, "Selectively hard negative mining for alleviating gradient vanishing in image-text matching," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 2, pp. 1921–1935, Feb. 2025.
- [59] Z. Li, C. Guo, Z. Feng, J.-N. Hwang, and Z. Du, "Integrating language guidance into image-text matching for correcting false negatives," *IEEE Trans. Multimedia*, vol. 26, pp. 103–116, 2024.
- [60] H. Diao, Y. Zhang, S. Gao, X. Ruan, and H. Lu, "Deep boosting learning: A brand-new cooperative approach for image-text matching," *IEEE Trans. Image Process.*, vol. 33, pp. 3341–3352, 2024.
- [61] H. Diao, Y. Zhang, L. Ma, and H. Lu, "Similarity reasoning and filtration for image-text matching," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2021, vol. 35, no. 2, pp. 1218–1226.



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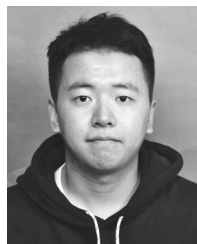
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